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I 003 LOCAL MOUNTED INSTRUMENT PANELS

1. SCOPE OF WORKS

The Technical Specification defines the general requirements for Local Mounted Instrument Panels to be supplied by the Contractor.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Sub-section 2.5 Codes and Standards”. The following Codes and Standards are extracted from Part 4, Sub-section 2.5 and specifically applicable to design, manufacturing, and testing of the equipment included in the specification.

ANSI B31.1	Code for pressure Piping
IEC 60079	Electrical apparatus for explosive gas atmosphere
IEC 60083	Plug and socket-outlet for domestic and similar general use
IEC 60529	Classification of degree of protection provided by enclosures
NFPA 70 Article 500	National Fire Protection Association (National Electrical Code), Hazardous (classified) location

3. ENVIRONMENTAL REQUIREMENTS

Local mounted instrument panels or racks shall be suitable for operation in a combined-cycle power plant near the sea coast. Refer to Part 4, Subsections 1.1, 2.14 and 2.16 for detailed information on meteorological, seismic and environmental conditions at the site.

4. DETAILED REQUIREMENTS

4.1 General

Local Mounted Instrument Panels shall be designed to accommodate local instruments, which are grouped together for installation and maintenance convenience. Such instruments shall include but not be limited to:

- Transmitters
- Transducers
- Non-indicating control devices
- Pressure and Temperature Indicators which cannot be pipe mounted
- Solenoid Valves not associated with control valves etc.

Instruments and control devices required to provide operator interface for controlling local systems shall not be mounted in Local Mounted Instrument Panels. Such instruments and control devices shall be mounted in control panels as specified in Part 4, Technical Specification I-002 “Local Control Panels”.

Local Mounted Instrument Panels shall be shop fabricated with all instruments mounted, piped to bulk head fittings, and wired to junction boxes as herein specified whenever possible.

If shop fabrication is not possible for reasons of scheduling and lack of timely information, the Contractor may, with prior approval by the Owner and evidence of suitable facilities, fabricate Local Mounted Instrument Panels at the job site.

Instruments, which for reason of size, weight or delicate nature cannot be safely shipped while mounted in a Local Mounted Instrument Panel, may be shipped separately and mounted in the panel at the job site. The mounting requirements and piping and wiring for such instruments shall be carried out in the shop prior to shipping as far as possible.

4.2 Control Power Supply

230 V, 50 Hz (nominal) power supply shall be provided for all Local Mounted Instrument Panels, where such a power supply is required.

All control circuits, control transformers and fuses shall be in accordance with Part 4, Technical Specification E-001 “General Electrical Specification.”

4.3 Construction

Each panel shall be a rigid, free standing structure, suitably braced, stiffened and reinforced to withstand all stresses incidental to shipping, installation and operation without warping or twisting greater than 6 mm as measured with a straight edge over the greatest straight line length across the panel section. The panel structure shall be fabricated of not less thick channel, angle or plate of hot or cold rolled steel. Welded construction shall be used throughout. Removable lifting eye bolts or angle and a dust and water protection cover shall be provided.

Internal brackets shall be provided for mounting accessory equipment. Structural steel framework shall be provided for supporting incoming cables, wiring, piping, and other appurtenance.

Instrument devices exerting strong cantilever forces on the panel face shall be adequately supported to remain plumb and true within 1.5 mm of vertical and horizontal alignment. Where devices are too large to be adequately supported for shipment to the Site and necessitate partial disassembly of the equipment, junction boxes shall be furnished and installed at all points where conduit is disconnected. Terminal blocks shall be furnished in these boxes. The contractor shall furnish all necessary flexible conduit and fittings to connect between junction boxes.

Each device shall be so mounted and piped with adequate spacing allowed for wiring, so that instrument removal, replacement and servicing may be accomplished without interruption of service to adjacent devices or unduly disturbing piping and tubing. All devices mounted inside the panels shall be so located that terminals and adjustment devices are readily accessible without the use of special tools and with terminal marking clearly visible.

In so far as is practical, the arrangement of instruments and conduits on panels shall place the electrical devices out of likely paths of condensate or water drains such as would occur when testing or calibrating instruments.

Indicating instruments shall be mounted with center lines at least 760 mm but not more than 1700 mm above the floor level.

Shut off, manifold and blow-down valves shall be provided as required and identified by the numbers of the instruments they serve.

Electrical relays shall be installed in hinged type enclosure separate from junction boxes with a degree of protection in accordance with Part 4, Technical Specification I-002. All spare relay contacts shall be wired to terminals in the junction boxes.

Panel or rack shipping length shall not exceed 2.5 m without prior approval by the Owner.

Local mounted instrument panels shall be **by contractor approval by the Owner.**

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4.4 **Painting and Finishing**

The Contractor's procedure for preparing the panel painting and the number and method of applications shall be in accordance with Part 4, Technical Specification I-002.

The contractor shall propose the colour of exterior surface to the Owner after contract awards for Owner approval.

4.5 **Name Plates**

Nameplates shall be installed to designate the service of all instruments in addition to any standard nameplate included with the instruments. A nameplate shall also be provided for each panel, designating its equipment number. The nameplate shall be in accordance with Part 4, Technical Specification I-002.

4.6 **Junction Boxes**

Two (2) junction boxes with a degree of protection of IPW 65 in accordance to IEC 60529 shall be furnished for each panel. The one junction box shall be used for the termination of "low level analog signals from the transmitters". The other junction box shall be used for the termination of "on-off contact-making signals (switches, relays, etc), and housing of fuses and indicating lights", when applicable.

The junction boxes shall be arranged to allow straight conduit entry to both boxes from the top or bottom. One (1) junction box per panel shall contain a pocket on the inside of the cover to hold a copy of the panel-wiring diagram. A minimum of 20 % spare terminals per junction box shall be provided.

Terminals associated with individual items of equipment shall be grouped and terminated at separate terminals for convenient cable connections. All spare contacts from auxiliary relays and switches shall be wired to terminals in the junction boxes as far as required.

One terminal shall be provided for each wire of the field cables. Any common connection required shall be provided by the contractor on the panel side of the block. A maximum of 2 conductors shall be connected to any one terminal post.

For shielded field cable, a separate terminal shall be provided for each shield drain wire.

Terminals shall be in accordance with Part 4, Technical Specification I-002.

Any nonmetallic components and devices such as terminals blocks, raceways, wire cleats, cable ties, etc., shall be manufactured from “non burning” materials.

4.7 Wiring

Wiring shall be in accordance with Part 4, Technical Specification I-002.

4.7.1 Arrangement of Wiring

Wiring on the panels between the junction boxes and the instruments shall be enclosed in watertight, flexible conduit.

All wiring between devices shall be continuous and splices made with wire nuts or similar devices are not acceptable. The conduit shall be connected to the wiring troughs on the junction boxes. The junction boxes shall be provided with drainage holes to allow breathing and drainage of condensate. All conduits shall be neatly located, grouped together where practical and substantially supported along the panel.

All wiring including spare wires shall be terminated at junction box terminal.

Wiring between transmitters and the signal junction box shall be run in separate conduits and raceways from that used for control and power wiring.

The wire insulation shall be removed without nicking or cutting the conductor. The wire shall be firmly inserted into the lug and crimped with a ratchet type or positive action air/hydraulic tool.

All wires or wire terminals projecting beyond the terminal block shall be insulated.

All bare bus and power connections shall be shielded to prevent accidental contact by persons.

4.8 Convenience Plug Receptacles

A two (2) pole, 15 Amp plug receptacle meeting the requirements for design C2a of IEC 60083 directly connected to the 220 VAC supply via suitable fuses shall be furnished on each panel. The plug receptacle shall be mounted in the power junction box.

4.9 Grounding

A grounding lug shall be attached to the main frame for receiving the 16 mm² bare, stranded copper ground cable.

The cases of all instruments shall be grounded.

Instrument circuits shall be grounded in accordance with the manufacture's instructions.

A stranded cable of 6 mm² shall be connected from a terminal point in each junction box to the panel ground lug. All wires in the junction boxes, which require grounding, shall be connected to these terminal joints.

4.10 Piping and Tubing

4.10.1 General

All instruments piping, tubing, valves and fittings shall be installed and securely fastened to the panels as required, grouped in an orderly manner and terminated with fitting and plugs at points located at the top of each panel. Each terminal point shall be identified by a stainless steel tag bearing the number of the associated instrument. All clamps, pads and supports in direct contact with stainless steel pipe or tubing shall be of stainless steel materials.

Piping or tubing shall not be routed across the face or rear of an instrument, junction box or other device in such a manner as to prevent the opening or removal of covers or to obstruct access to leads, terminals or instruments for servicing.

Test connections shall be provided so that each instrument can be calibrated and checked without its removal from the panel.

All valves located within runs (tube connection at inlet and outlet) shall be mechanically supported to the panel structure. Valves in piping runs may be supported by the pipe, provided adequate pipe supports are utilized.

Piping and tubing runs shall be continuous from equipment to the terminal area except where tees are required or where tubing cannot properly be formed, attached or removed in one piece without deformation. The arrangement shall be logical, orderly, accessible and neat with straight, parallel runs and a mini-

mum of crossing. Vertical run shall be plumb. Horizontal runs shall pitch slightly away from the instrument. Pneumatic tubing shall be kept segregated from process piping.

Supports shall provide complete freedom from strain on equipment, and runs shall be so arranged that connections can be broken without distortion of the tubing.

4.10.2 Process Sensing Lines

All process sensing line piping, fitting and welding within the panels, shall confirm to the ANSI B 31.1 code for power piping. Contractor shall certify that the design, materials, fabrication, construction and testing of all piping provided by him conform to ANSI B 31.1 code.

All process-sensing lines fittings and connections shall be of welded constructions except final connections to the instruments and test connections at the instruments which shall be threaded.

4.10.3 Instruments Air

The contractor shall connect instrument air to those panels which require it. Panels having 3 or more air users shall be provided with a 12 mm air header. The contractor shall furnish air filter pressure regulators where required.

Tubing bend radius shall be 5 diameters minimum unless otherwise specified.

Connections from control device shall be terminated in bulkhead fittings on horizontal strap-type brackets located at the top of the panel. All signal output and input connections shall be furnished with a tee and plugged shutoff valve for local test reading of instrument signals.

Each air supply stop valve and tube terminals connections shall be identified by a stainless steel tag stamped with tag number of the associated instruments.

One spare-plugged air supply shutoff valve for instruments testing shall be provided on each panel.

4.11 Hazardous Location

Instrumentation and control devices and equipment located in hazardous location (area) shall be in accordance with NFPA 70, Article 500 and IEC 60079.

5. TESTING

5.1 Electrical Test

The electrical test shall demonstrate the freedom from grounds and short circuits and the accuracy of wiring of all panel-mounted devices. The tests shall include point-to-point, continuity and electrical insulation test in accordance with the applicable ANSI Standard (Note: Megger or other high voltage shall not be applied to any coaxial shielded cable or solid-state equipment).

The wiring shall be capable of withstanding for one (1) minute field test at 1500 VAC, between all parts to ground.

The Contractor shall be responsible for proper protection of instruments and devices that may be damaged by high voltage tests and checks shall be made by the Contractor before shipment to ensure that all electrical equipment and wiring furnished and installed by the Contractor is in proper operating condition and that the wiring is in exact accordance with the wiring diagrams and that all wiring will function as intended.

All wiring for instruments shipped separately or at a later date than the panel shall be completed and tested.

The expense of any necessary reworking of the wiring shall be the responsibility of the Contractor.

5.2 Hydrostatic Test

Demineralized water shall be used for all hydrostatic testing, and the test shall be accordance with the requirements of ANSI B 31.1.

All tubing and piping connections, welds and valve stems and seats shall be checked for leaks. Detectable leakage shall not be permitted. Leaks shall be repaired and the equipments retested.

During the hydrostatic test, all instruments and devices shall be disconnected beyond the shutoff valves. Valves shall be placed in an open condition and the line or valve capped. This is intended to test the valve bodies only and prevent damage to the valve seals.

5.3 Pneumatic Test

All pneumatic piping, tubing, valve and fittings shall be tested with 10.5 bar nitrogen in accordance with the requirements of ANSI B31.1, with the final connection at the instruments disconnected and plugged. Each joint shall be inspected for leaks using a soap bubble-type test (chlorine or chloride free fluid). Detectable leaked shall not be permitted. Leaks shall be repaired and re-tested.

6. QUALITY ASSURANCE QUALITY CONTROL (QA/QC)

Quality Assurance/Quality Control shall conform to the requirement of Part 4, Sub-section 2.7.

The Quality Control procedures as specified Part 4, Sub-section 2.7 shall included, but not be limited to the following:

- Continuity and Wiring Tests
- Insulation and High Potential Testing
- Packaging and Shipping
- Welding
- Cleaning and Painting

Quality Assurance/Quality Control documentation as specified in Part 4, Sub-section 2.7 shall included, but not be limited to the following:

- ShopTest Reports
- All other documentation required by application codes and standards

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